

Agrumi-Gel s.r.l.	TECHINAL DATA SHEET	Cod.: D 10.03.68 EN Rev.: 12 Date: 01/08/2023 Issued by: Elsa Liga Approved by: Salvatore Rizzo Revision Reason: Update Pag. 1 di 7
-------------------	----------------------------	---

Name of product:	Mandarin juice concentrate 60° brix frozen
Description:	Mandarin juice concentrate 60° brix frozen
Identification of batch:	CMS

1. Composition

Ingredients: Mandarin (100%)

Identification of the ingredient and guidance on the supply

Raw material: Mandarin

Botanical name: Citrus reticulata

Country of origin: Sicily (Italy)

Variety: mandarin common, late from Ciaculli

Harvesting period: Typically from January to May

Determination of maturity: Brix, Acidity, Color (pigmentation)

Packaging: Plastic bins of 264 Kg. $\pm$ 14

Addition: 0.007 % of essential oil mandarin

On the raw material there hasn't been carried out microbiological tests, chemical test aimed to detect contaminants nor panel test.

Agrumi-Gel s.r.l.	TECHINAL DATA SHEET	Cod.: D 10.03.68 EN Rev.: 12 Date: 01/08/2023 Issued by: Elsa Liga Approved by: Salvatore Rizzo Revision Reason: Update Pag. 2 di 7
-------------------	----------------------------	---

2. Physical, organoleptic, chemical and microbiological parameters

Chemical-Physical Parameters

Parameters	Target	Min	Max	Unit	Method
Brix	60	57	63	°	IFU 8,1991
Acidity	5.5	5	6	% p/p	IFU 3,1996
Ratio		11.2	13		°brix corrected/acidità
Ascorbic acid	>400	300	800	Mg/l	Internal method
Pulps at 1500		2	4	%v/v	IFU 60,1991
Pulps at 3000		1	2	%v/v	IFU 60,1991

* in relation to the season

Microbiologic Parameters

Organism	Target	Min	Max	Unit	Method
CMT	-	-	100	cfu/ml	Compact Dry TC 48 ± 3 hours / 30 ± 2 °C
Yeasts	-	-	10	cfu/ml	Compact Dry YM 3-7 days / 25 - 30 °C
Moulds	-	-	10	cfu/ml	Compact Dry YM 3-7 days / 25 - 30 °C
Total Coliforms	Absent	-	-	cfu/ml	Compact Dry EC 24 ± 2 hours / 37 ± 1 °C
E. Coli	Absent	-	-	cfu/ml	Compact Dry EC 24 ± 2 hours / 37 ± 1 °C
Salmonella	Absent	-	-	cfu/ml	Compact Dry SL 20-24 hours / 36 ± 1 °C

Organoleptic parameters

Characteristics	Target	Method
General Aspect	Liquid	Panel Test
Color	Yellow-orange	Panel Test
Taste / Aroma	Characteristic of Mandarin	Panel Test
Sensation to the palate	Characteristic of Mandarin	Panel Test

Agrumi-Gel s.r.l.	TECHINAL DATA SHEET	Cod.: D 10.03.68 EN Rev.: 12 Date: 01/08/2023 Issued by: Elsa Liga Approved by: Salvatore Rizzo Revision Reason: Update Pag. 3 di 7
-------------------	----------------------------	---

3. Legal requirements for food safety product

- Directive 2012/12/UE and subsequent amendments of the European Parliament and of the Council of 19 April 2012 amending the Directive 2001/112/CE of Council relating to fruit juice and certain similar products intended for human consumption;
- Directive 2003/89/CE and subsequent amendments of the European Parliament and of the Council of 10 November 2003 amending the Directive 2000/13/CE relating to the labeling, presentation and advertising of foodstuff;
- Directive (UE) 2015/1787 and subsequent amendments of the Commission of the 06 October 2015 amending Annexes II and III of the Directive 98/83/EC of the Council on the quality of water intended for human consumption;
- Directive 2001/111/CE and subsequent amendments of 20 December 2001 relating to certain sugar intended for human consumption;
- Directive 2014/63/EU and subsequent amendments of the European Parliament and of the Council of 15 May 2014 amending Council Directive 2001/110/EC relating to honey;
- Regulation (EC) n° 1333/2008 and subsequent amendments of the European Parliament and of the Council of 16 December 2008 on food additives;
- Commission Regulation (EC) n° 1130/2011 and subsequent amendments of 11 November 2011 amending Regulation (EU) n° 1333/2008 of the European Parliament and of the Council on food additives by establishing a Union list of food additives approved for use in food additives, food enzymes, food flavorings and nutrients;
- Commission Regulation (EU) n° 231/2012 and subsequent amendments of 9 March 2012 laying down specifications for food additives listed in Annex II and III of Regulation (EC) n° 1333/2008 of the European Parliaments and of the Council;
- Commission Regulation (EU) n° 257/2010 and subsequent amendments of 25 March 2010 establishing a program for the re-evaluation of approved food additives in accordance with Regulation (EC) n° 1333/2008 of the European Parliament and of the Council;
- The product must be considered "not genetically modified" or "not derived from a genetically modified organism" as defined by the EC 1830/2003/EC on labeling and traceability and the EC directives 1829/2003/EC on foods and genetically modified feeds and subsequent amendments;
- The product does not contain substances that have allergenic properties for which labeling is required as defined as in Regulation (UE) n.1169/2011 and subsequent amendments of the European Parliament and of the Council of the 25 October 2011;
- The product was not produced and does not contain any ionized product in reference to the Directive EU 1992/2/EC and 1999/3/EC and subsequent amendments;

Agrumi-Gel s.r.l.	TECHINAL DATA SHEET	Cod.: D 10.03.68 EN Rev.: 12 Date: 01/08/2023 Issued by: Elsa Liga Approved by: Salvatore Rizzo Revision Reason: Update Pag. 4 di 7
-------------------	----------------------------	---

4. Treatment methods

Flow-chart

D 09.01 01 – D 09.01 13

Phases of process

Receiving Citrus Fruit and Unloading – Washing – Selection and Calibration –

Juice Extraction – Refining – Refrigeration - Temporary storage - Preheating –
 Refining – Centrifugation - Sending to the Finished Products Department –
 Pasteurization - Refrigeration – Filtration and magnetic trap - Packing – Freezing –
 Storage – Loading on the truck or in tanker

5. Packaging

On customer request

Metal conical drums

Gross weight	264 Kg	
Net weight	250 Kg. $\pm$ 1 %	
External packaging	Metal conical drums (NEW)	<i>The drums are sealed and the internal paint is food. The cover is of metal and sealed with a metal ring in turn sealed with a plastic seal that shows the possible tampering. The drum is free of rust, painted uniformly, free of excessive bruising. The steel drums are in accordance with the M.D. 21.03.1973 and subsequent updates, the heavy metals content is less than 100 ppm in accordance with the Directive 94/62/EEC and have been produced in accordance with the good manufacturing practices as per the regulations 1935/2004/EC and 10/2011/EC. The drums design is hermetic and the suppliers tests confirm this requirement. The material respects the limits of global migration and specific migration of the dye, therefore, it can be used in prolonged contact with foods.</i>
Internal packaging	Double polyethylene and transparent bag for food	<i>All inner bags are new and compounds of low density polyethylene material whose components are approved for food use. The bags are suitable to come into contact with foodstuffs according to the M.D. 21.03.1973. the bags have a minimum thickness of 50 microns. The material respects the limits of global migration and specific migration of the dye, therefore, it can be used in prolonged contact with foods.</i>
Seal type	N° 2 numbered plastic seals, one per each batch N° 1 not numbered plastic seal for the drum's cover	

Possible variation on demand: Grey metal cylindrical drums, blue or green metal conical drums, blue polyethylene bag, numbered plastic seal for the drums cover, numbered metal seals.

Plastic pails

Gross weight	26 Kg.	
--------------	--------	--

Agrumi-Gel s.r.l.	TECHINAL DATA SHEET	Cod.: D 10.03.68 EN Rev.: 12 Date: 01/08/2023 Issued by: Elsa Liga Approved by: Salvatore Rizzo Revision Reason: Update Pag. 5 di 7
-------------------	----------------------------	---

Net weight	25 Kg. $\pm$ 1 %	
Packaging	Plastic pails (NEW)	<i>Plastic pails, as per the technical sheet provided to us by the supplier, are produced with materials and dyes suitable for food contact and meet the levels of concentration of heavy metals in terms of packaging. The material respects the limit of global migration and specific migration of the dye, therefore, it can be used in prolonged contact with foods.</i>
Seal type	Lid pressure closure with a warranty seal	

Other packaging's

On customer request the product may also be packed in Octabins or aseptic bags (in drums or in carton boxes).

5a Labeling

The external package is supplied with a label giving at least the following information:

- ✓ Product name (with indication of origin coming from organic farming):
- ✓ Indication of batch:
- ✓ Ingredients:
- ✓ Producer name:
- ✓ Production date:
- ✓ Packaging date:
- ✓ Expiry date:
- ✓ Net Weight (Kg.):
- ✓ Gross weight (Kg.):
- ✓ Storage conditions:

On customer request and approval by Agrumi-Gel S.r.l. the label may bear more information.

General Notes

The product in drums may also be supplied without pallets. The final products supplied on pallets are secured by strapping and wrapping material. The pallets are in good conditions, they are free from splinters and free of obvious signs of moisture and molds. They are free of protruding or easily removable nails.

To ensure the maximum safety during all phases of transportation and handling, all packaging are suitable for food and specifically designed for its intended use.

For each type of packaging a system is provided to ensure that the product cannot be tampered without a clear evidence of tampering.

For each type of package on request, there is provided, the data sheet of the supplier.

All packages have passed a careful analysis of risks within the HACCP plan.

All packages are stored in order to minimize the risk of contamination and spoilage.

Tankers

The product can also be supplied in tanker. For this type of transport the requirements are, besides a purchase order, the respect of following conditions:

- *The vehicle must exclusively be used for food transportation and must have a clearly visible labeling indissolubly fixed to specify it; the parts which are in contact with the product may not have any*

Agrumi-Gel s.r.l.	TECHINAL DATA SHEET	Cod.: D 10.03.68 EN Rev.: 12 Date: 01/08/2023 Issued by: Elsa Liga Approved by: Salvatore Rizzo Revision Reason: Update Pag. 6 di 7
-------------------	----------------------------	---

negative effect on the product (chemical, physical, sensorial or hygienic); the vehicle of transport and all the piping systems must be easy to clean and disinfect, and restored;

- *The transported products and so the products designated for cleaning and restoration of the vehicle must be completely and easily unloaded from the outlet valve DN 65 – for low viscosity products; and from outlet valve DN 80 and DN 100 for medium-high viscosity products;*
- *The tanker shall have a degree of insulation such as to limit the increase in temperature of the product inside the MAX 1° C/day; the driver of the tanker must have the possibility to control the temperature;*
- *The tanker must have openings which allow an easy access for sampling;*
- *The tanker must be provided with relief valves equipped with a special filter to prevent intrusion of foreign bodies;*
- *The tanker must be equipped with ladders and safety railing and in good condition; the possibility to sterilize the gaskets must be given;*
- *During the loading and unloading of the delivery an appropriate retina anti-insect must be applied on the openings of the tanker to avoid any possible intrusion of foreign bodies;*
- *The tanker must be conform to all legal norms and may not show signs of damage nor sign of poor maintenance.*

For the collection of the delivery the driver must show up at the indicated place on site with an original cleaning/washing certificate. The washing certificate may not be issued more than 48 hours before the loading.

The certificate must include: the number of seals placed to the closing filler; indication of the products of the 3 (three) previous loads ; the name of the carrier; date and time of beginning and ending of the cleaning process; the type of cleaning and the products which were applied; details of the cleaned parts; stamp and signature of the washing station.

All cleaning agents and also of disinfection must be approved for food use and the water must be potable (Quality standard EEC 98/83 and updates).

Tankers which have a multi camera system with connecting valve must be cleaned and sterilized completely even though they were used partially. The washing process in sectors may only be allowed if the air pipelines and connecting valves are completely separate.

5b Unauthorized previous cargoes

GMO products and products covered by the Directive 2003/1829/EC and subsequent amendments and additions.

Allergens and products covered by the Directive 2003/89/EC and subsequent amendments and additions.

6. Shelf-Life

36 months from date of production; proper storage and packaging integrity; at temperature of -20 °C ± 2 (for Frozen Packaged product).

7 days from date of production at a temperature of +0 °C ± 2 (for the product refrigerated in tanker).

Agrumi-Gel s.r.l.	TECHINAL DATA SHEET	Cod.: D 10.03.68 EN Rev.: 12 Date: 01/08/2023 Issued by: Elsa Liga Approved by: Salvatore Rizzo Revision Reason: Update Pag. 7 di 7
-------------------	----------------------------	---

7. Storage conditions and shipping method

The product should be stored in cold storage room and transported on refrigerated truck at a temperature of $-20^{\circ}\text{C} \pm 2$ (for Frozen Packaged product).

The product must be transported in tanker at a temperature of $0^{\circ}\text{C} \pm 2$ (for the product Refrigerated in Tanker).

8. Intended and expected use

Semi-finished product to be used for the food industry.

Agrumi-Gel s.r.l. (the "supplier") is committed to provide the product according to the specifications given in the data sheet with the exception of the parameters of product or process otherwise specified by the customer in writing. In relation to the particular conditions of the citrus season, upon notice before signing the contract, the supplier reserves the right to modify certain parameters in the data sheet. The customer, upon signing the contract which contains the identifier of the technical details, declares to accept the parameters of process and product in itself contained with the exception of the different parameters required, according to the above conditions, and expressly accepted by the supplier.